

Exercise 6.1 a-

Question No 1 a-

18 = mathematics

325 = Computer science

(a)

Product : (x)

sum : (+)

Independent choice

In how many ways two representative be picked so that one is a mathematical major and other one is Computer Science major?

$$18 \times 325 \\ = \boxed{5850}$$

(b)

Single choice
~~either~~ ~~and~~

In how many ways can one representative be picked who is either a mathematical or a Computer Science major.

$$18 + 325 \\ = \boxed{343}$$

Question No 2 a-

Building contain 27 floors

37 offices on each floor

How many offices are in the building?

$$= 27 \times 37$$

$$= \boxed{999}$$

Question No 3 :-

Four multiple choices 10,
Four possible answers.

(a)

In how many ways can a student
ans the question on the test if
the student ans every questions?

$$= 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4$$

$$= \boxed{4^{10}}$$

(b)

Ans: the student on the test if
the student can leave the blank?

4 leave

1 own option

Now

$$= \boxed{5^{10}}$$

Question No 4 :-

Colour Option = 12

Version male and female = 2

Sex size = 3

How many different types of this shirt
are made?

$$= 12 \times 3 \times 2$$

$$= \boxed{72}$$

Question No 5 :-

Six fly from New York to Denver.

seven fly Denver to San Francisco

$$= 6 \times 7$$
$$= \boxed{42}.$$

Question No 6:-

Four major autoroutes from Boston to Detroit.

Six from Detroit to Los Angeles

$$= 4 \times 6$$
$$= \boxed{24}.$$

Question No 7:-

How many different three letters initials can people have?

Sol:-

$$= 26 \times 26 \times 26$$

$$= 26^3$$

Question No 8:-

How many different three-letters initials with none of the letters repeated can people have?

$$26 \times 25 \times 24$$

Question No 9:-

How many different letters are ^{three} that Begin with an A?

1 letter begin with A
The next two letters
 26×26

$$26^2$$

Question 1010 :- 2

How many bit strings are there
of length eight?

Soln:-

$$\frac{01}{2}$$

$$2^8$$

Question 1011 :-

How many bit strings of length
ten both ^{start} begin and ^{end} end with a 1?

start and end

1

1

= 2

Remaining 8

$$2^8$$

Question 1012 :-

How many bit strings are there
of length six or less, not counting the
empty string?

$$= 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6$$

$$= 126$$

$$\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{matrix}$$

Question No 13 :-

How many bit string with length not exceeding n , whenever n is a positive integer, consists entirely of 1's not counting the empty string.

Sol :-

$$n+1$$

Question No 14 :-

How many bit strings of length n , where n is a positive integer, start and end with 1's?

$$n-2 \quad \text{Ans}$$

Example :- Fixed

$$1 \text{ (1) } 1 \text{ (1) } 1 \text{ (1) } 1 \text{ (1) } 1 \text{ (1) } 1 \text{ (1)}$$

Question No 15 :-

How many strings are there of lowercase letters of length four or less, not counting the empty string?

$$\begin{aligned} &= 26^1 + 26^2 + 26^3 + 26^4 \\ &= 26 + 676 + 1756 + 456976 \\ &= 475,254 \end{aligned}$$

Question No 16 :-

How many strings are there of four lowercase letters that have the letter x in them?

$$\begin{aligned} &= 26^4 - 25^4 \\ &= 66351 \end{aligned}$$

Question No 17:-

How many strings of five ASCII characters contain the character '@' ("at" sign) at least once? (Note: There are 128 different ASCII characters.)

$$= 128^5 - 127^5$$

RNA = AUG

Question 19.

How many 6 element RNA sequences

(a)

Do not contain U _____ ?

ACG ACG ACG ACG ACG ACG ACG

$$= 36$$

(b)

end with GV?

AUCC AUCC AUCC AUCC AUCC AUCC

$$= 4$$

(c) + 1

Start with C ?

AUCG AUCG AUCG AUCG AUCG AUCG
Start. []

$$= 45$$

(d)

Contain only A or U?

AV AV AV AV AV AV AV

$$= 2^6$$

DNA = ACTG

Question no 18:-

How many 5-element DNA sequences

ACTG ACTG ACTG ACTG ACTG

(a)

end with A?

ACTG

One position fix, remaining 4

$$= 4^4$$

(b)

Start with T and end with G?

ACTG ACTG ACTG ACTG ACTG

Start T

End G

$$= 4^3$$

(c)

Contain only A and T?

AT AT AT AT AT AT

$$= 2^5$$

(d)

do not contain C?

ATG ATG ATG ATG ATG ATG

$$= 3^5$$

Question no 20:-

How many positive integers between 5 and

31.

Table of 3:- (a)

are divisible by 3? which integers are these?

6, 9, 12, 15, 18, 21, 24, 27, 30

Table of 4. (b)

are divisible by 4? which integers are these?

8, 12, 16, 20, 24, 28.

(c)

are divisible by 3 and 4? which integers are these?

12, 24

Question 21 e-

How many positive integers b/w 50 and 100?

Tables

(a)

are divisible by 7? which integers are these?

56, 63, 70, 77, 84, 91, 98.

(b)

are divisible by 11? which integers are these?

55, 56, 77, 88, 99.

(c)

are divisible by both 7 and 11? which integers are these?

77.

Question No 22 :-

How many positive integers less than 1000

(a)
are divisible by 7?

Total 142 numbers.

(b)
are divisible by 7 but not 11?

Total 130 numbers

(c)
are divisible by both 7 and 11?

= 12 multiples of 77.

(d)
are divisible by either 7 or 11?

= 220

(e)
are divisible by exactly one of 7 and 11?

= 208

(f)

are divisible by neither 7 nor 11?

$$= 779$$

(g)

have distinct digits?

1 digit number equal to 9

2 digit number equal to 81

3 digit number equal to 648

(h)

$$9 + 81 + 648 = \boxed{738}$$

have distinct digits and are even?

1 digit number equal to 5

2 digit number equal to 40

3 digit number equal to 320

$$= 5 + 40 + 320$$

$$= \boxed{365}$$

Question No 23:-

How many positive integers between 100 and 999 inclusive.

(a)

are divisible by 7?

~~128~~ Total $\rightarrow$ 128

(b)

are odd?

$$= 450 \text{ Total}$$

(c)

have the same three decimal digits?

$$\text{Total} = 9$$

(d)

are not divisible by 4?

$$\text{Total} = 675$$

(e)

are divisible by 3 or 4?

$$\text{Total} = 450$$

(f)

are not divisible by either 3 or 4?

$$\text{Total} = 450$$

(g)

are divisible by 3 but not by 4?

$$\text{Total} = 225$$

(h)

are divisible by 3 and 4?

$$\text{Total} = 75000$$

Question No 24 :-

How many positive integers between 1000 and 9999 inclusive.

(a)

are divisible by 9?

$$1000$$

(b)

are even?

$$4500$$

(c)

have distinct digits?

$$4536$$

(d)

are not divisible by 3?

$$= 6000$$

(e)

are divisible by 5 or 7?

$$= 2829$$

(p)
are not divisible by either 5 or 7?
= 6171

(q)
are divisible by 5 but not by 7?
= 1543

(h)
are divisible by 5 and 7?
= 257

Question → How many strings of three decimal digits:
 $\overset{0}{\underset{9}{\text{---}}} \quad \overset{0}{\text{---}} \quad \overset{0}{\text{---}}$ $\overset{10^3}{\text{---}} \quad \overset{0}{\text{---}} \quad \overset{9}{\text{---}}$

(a)
do not contain the same digit three times?

012 345 678
 $10 \times 10 \times 10$
 $= 10^3 = 1000 - 10 = \boxed{990}$

(b)
begin with an odd digit?

odd integers 1, 3, 5, 7, 9
 $= 5 \times 10 \times 10 = \boxed{500}$
 ↳ decimal

have exactly two digits that are 4s?

Position = 3

$\textcircled{4}\textcircled{4}\textcircled{1}$ $\textcircled{4}\textcircled{4}\textcircled{2}$ $\textcircled{4}\textcircled{4}\textcircled{3}$

$= 3 \times 9 = \boxed{27}$

Question no 26 e.

How many strings of four decimal digits

do not contain the same digit twice?

$10^4 = 10000$
 $= 10 \times 9 \times 8 \times 7 = \boxed{5040}$

end with an even digit?

$0, 2, 4, 6, 8$
 $10 \times 10 \times 10 \times 5 = \boxed{5000}$

have exactly three digits that are 9s?

$\textcircled{9}\textcircled{9}\textcircled{9}\textcircled{1}$ $\textcircled{9}\textcircled{9}\textcircled{9}\textcircled{2}$ $\textcircled{9}\textcircled{9}\textcircled{9}\textcircled{3}$ $\textcircled{9}\textcircled{9}\textcircled{9}\textcircled{4}$

$= 4 \times 9 = \boxed{36}$

Question no 27 e.

Committee formed one representative from each of 56 states in United States. where the representative from a state is either the governor or one of the two senators from that state. How many ways are there to form this committee?

One Governor
two Senators
and states = 50
= 3^{50}

Question No 28 e.

How many license plates can be made using either three digits followed by three uppercase English letters or three uppercase English letters followed by three digits?

Statement 1:

$$= 10^3$$

$$\times 26^3$$

Statement 2:

$$= 26^3$$

$$\times 10^3$$

$$= 10^3 \times 26^3 + 26^3 \times 10^3$$

$$= 2 \times 10^3 \times 26^3$$

Question No 29.

How many license plates can be made using either two uppercase English letters followed by four digits or two digits followed by four uppercase English letters?

Statement 1:

$$26^2 \times 10^4$$

Statement 2:

$$10^2 \times 26^4$$

$$10^2 \times 26^4 + 26^2 \times 10^4$$

$$(26^2 \times 10^2) + (10^2 \times 26^4)$$

Question No 30:-

How many license plates can be made using either three uppercase English letters followed by three digits or four uppercase English letters followed by two digits?

Statement 1:-

$$26^3 \times 10^3$$

Statement 2:-

$$26^4 \times 10^2$$

$$(26^3 \times 10^3) + (26^4 \times 10^2)$$

Question No 31:-

How many license plates can be made using either two or three uppercase English letters followed by either two or three digits.

Possible statements:-

$$26^2 \times 10^2$$

$$26^3 \times 10^2$$

$$26^2 \times 10^3$$

$$26^3 \times 10^3$$

$$= (26^2 \times 10^2) + (26^3 \times 10^2) + (26^2 \times 10^3) + (26^3 \times 10^3)$$

Question no 32 a-

26^8

How many strings of eight uppercase English letters are there.

(a)

if letters can be repeated?

$$= 26^8$$

(b)

if no letters can be repeated?

length = 8

$$= 26 \times 25 \times 24 \times 23 \times 22 \times 21 \times 20 \times 19$$

(c)

that start with X, if letters can be repeated?

$$X = \text{Fix}$$

$$= 26^7$$

(d)

that start with X, if no letters can be repeated?

$$X = \text{Fix}$$

$$= 25 \times 24 \times 23 \times 22 \times 21 \times 20 \times 19$$

One fix

(e)

that start and end with X, if letters can be repeated?

$$\text{Fix } (26) \times 25 \times 24 \times 23 \times 22 \times 21 \times 20 \times (19)$$

$$= 26^6$$

(f) that start with the letters BO (in that order), if letters can be repeated?

$$= 26^6$$

(g) that start and end with the letters BO (in that order), if letters can be repeated?

$$25 \times 24 \times 23 \times 22 \times 21 \times 20$$

$$= 26^4$$

(h) that start or end with the letters BO (in that order), if letters can be repeated?

$$= 2 \times 26^6$$

Question No 33

How many strings of eight English letters are there.

(a)

that contain no vowels, if letters can be repeated?

a, e, i, o, u x

$$= 21^8 \text{ Ans}$$

(b)

that contain no vowels, if letters cannot be repeated?

$$2 \times 1 \times 20 \times 19 \times 18 \times 17 \times 16 \times 15 \times 14$$

(c)

that start with a vowel, if letters can be repeated?

@, e, i, o, u

one fix length and then

$$= 5 \times 26^7$$

(d)

that start with a vowel, if letters cannot be repeated?

$$5 \times 25 \times 24 \times 23 \times 22 \times 20 \times 21 \times 19$$

(e)

that contain at least one vowel, if letters can be repeated?

$$= 26^8 - 21^8$$

(f)

that contain exactly one vowel, if letters can be repeated?

One fix =

$$8 \times 5 \times 21^7$$

$$5 \times 21^7$$

that start with (g)

that start with x and contain at least one vowel, if letters can be repeated?

$$26^7 - 21^7$$

5

that start and end with x and contain at least one vowel, if letters can be repeated?

$$26^6 - 21^6$$

Question No 42

How many 4-elements DNA sequences

(a)

do not contain the base T?

ACG ACG ACG ACG

$$3^4$$

(b)

Contain the sequences ACG?

① A C G ②

ACG ACG ACG ACG

length = 4

$$2 \times 4 = 8$$

(c)

Contain all four bases A, T, C and G?

ATCG, ATCG, ATCG, ATCG

A T C G

$$4 \times 3 \times 2 \times 1 = 24$$

(d)

Contain exactly three of the four bases A, T, C, and G?

$$4 \times 3 \times 2 = 24$$

258 → 100
③

Question No 43 :-

How many 4-element RNA sequences

AUCG

(a)

Contain the base U?

U U U U

4 4 4 4

$$4^4 = 256$$

$$4^4 - 3^4$$

(b)

do not contain the sequence CUG?

A A A A

$$4^4 - 8 = 248$$

$$\therefore 2 \times 4 = 8$$

CUG can fix at

two positions

(c)

do not contain all four bases A, U, C and G?

$$4^4 = 256$$

$$4 \times 3 \times 2 \times 1 = 24$$

$$24 - 256 = 232$$

(d)

Contain exactly two of the four bases A, U, C and G?

Total number of sequences $2^4 = 16$

Removing one Base Sequence $16 - 2 = 14$

Multiplying by the number of base pairs $6 \times 14 = 84$

6-sequence
(A,U), (A,C), (A,G), (U,C),
(U,G), (C,G)

Question No 44 :-

How many ways are there to seat four of a group of ten people around a circular table where two seatings are considered

when everyone has the same immediate
left and immediate right neighbor?

4 group of ten people

2 seating